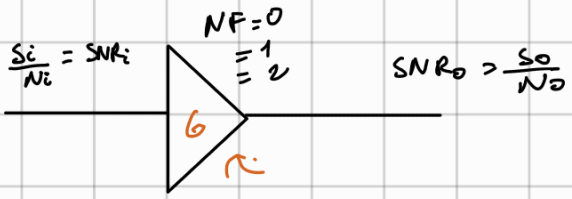
LNA



$\frac{S_i}{N_i}$

$N_o = N_i G + N_A$

$SNR_o < SNR_i$

$SNR_i > SNR_o$

Low Noise Figure $NF_{dB} = \frac{SNR_i}{SNR_o} = SNR_{[dB]} - SNR_o\,[dB]$

($\frac{SNR_i}{SNR_o} = F$)

$NF = 10 \log F$

F: Noise factor

NF: Noise figure

istenilenler

Linearity

Low power consumption

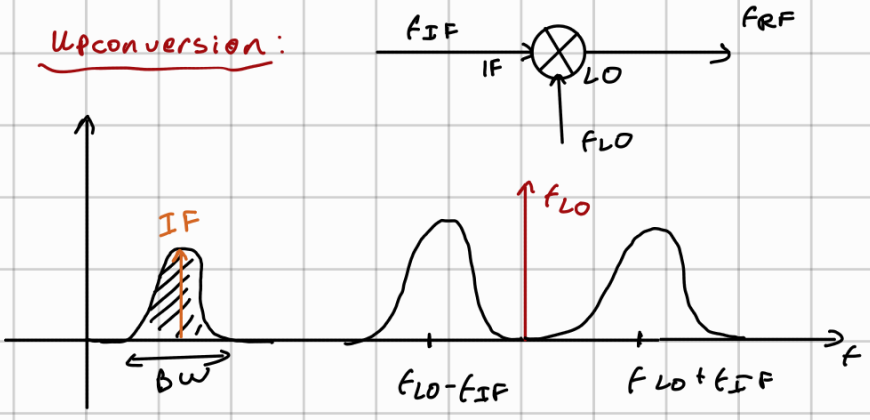
Low area "

Gain $[S_{21}]$

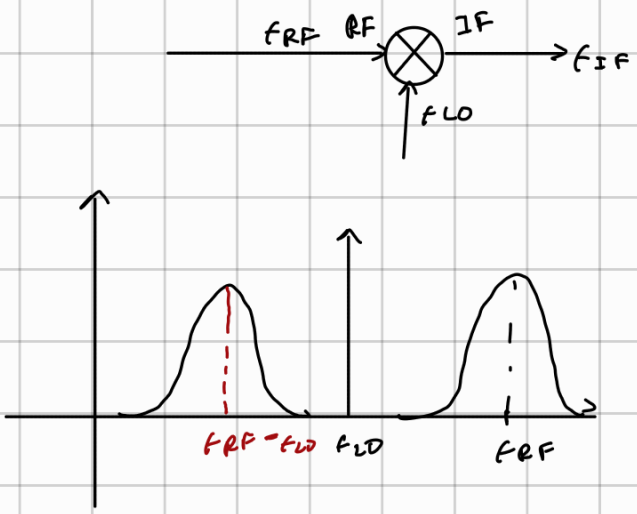
Matching

Mixers

Upconversion:



Down conversion



$f_{IF} = f_{RF} - f_{LO}$

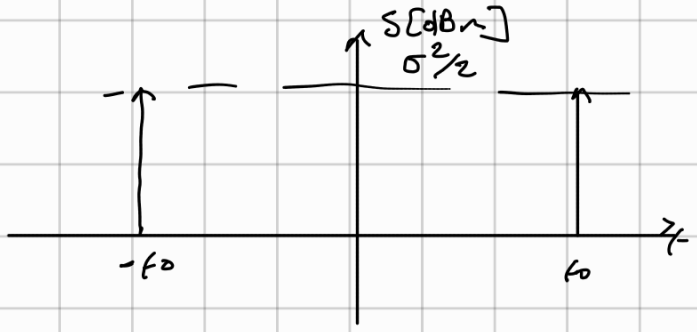
$$g(t) = \cos 2\pi f_0 t \longrightarrow G(f) = F\{g(t)\} = \int_{-\infty}^{\infty} \cos 2\pi f_0 t \; e^{-j2\pi f t}\, dt$$

$$= \int_{-\infty}^{\infty} \frac{e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}}{2} e^{-j2\pi f t}\, dt$$

$$= \frac{1}{2}\left\{ \int_{-\infty}^{\infty} e^{j2\pi f_0 t} e^{-j2\pi f t}\, dt + \int_{-\infty}^{\infty} e^{-j2\pi f_0 t} \cdot e^{-j2\pi f t}\, dt \right\}$$

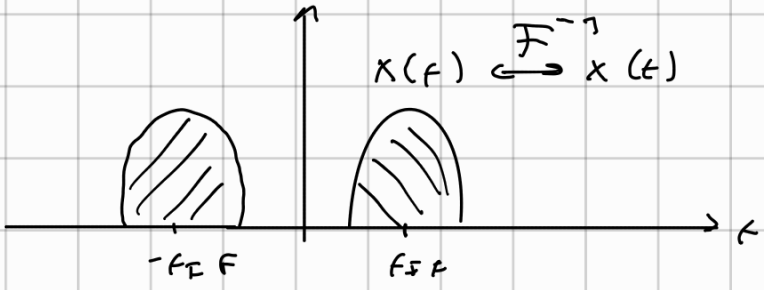
$$= \frac{1}{2}\left[\delta(f-f_0) + \delta(f+f_0)\right]$$





$g(t) = \cos 2\pi f_0 t$

$\sigma^2 = E[g^2(t)] - E^2[g(t)]$



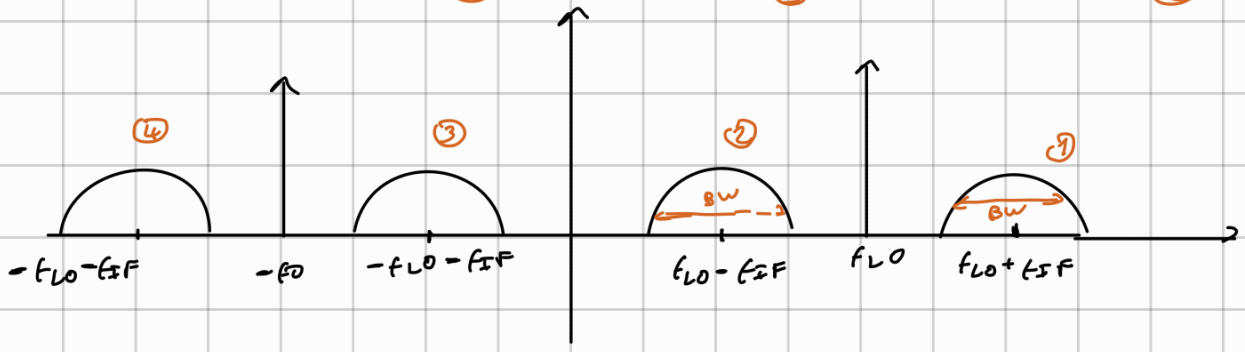
$$y_{RF}(t) = x(t)\left[\cos 2\pi f_{LO} t\right]$$

$$Y(f) = X(f) \circledast \frac{1}{2}\left[\delta(f-f_{LO}) + \delta(f+f_{LO})\right]$$

$$= \left[X(f-f_{IF}) + X(f+f_{IF})\right] \circledast \frac{1}{2}\left[\delta(f-f_{LO}) + \delta(f+f_{LO})\right]$$

$$= \frac{1}{2}\left[X(f-f_{IF}-f_{LO}) + X(f-f_{IF}+f_{LO}) + X(f+f_{IF}-f_{LO}) + X(f+f_{IF}+f_{LO})\right]$$

$f-[f_{IF}+f_{LO}]$ ① $\quad f-[f_{IF}-f_{LO}]$ ③ $\quad f-[f_{LO}-f_{IF}]$ ② $\quad f-[-f_{LO}-f_{IF}]$ ④

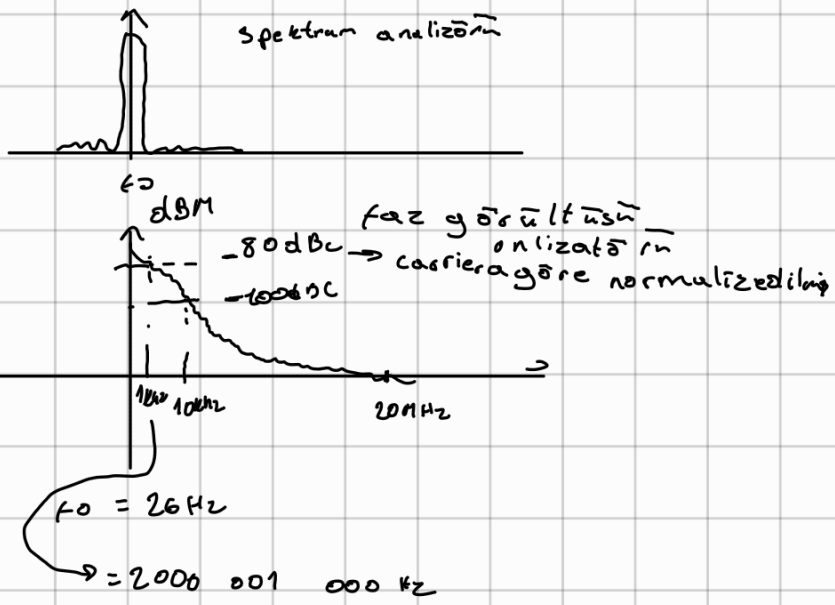


$x(t) = \cos 2\pi f_1 t \rightarrow f_1 = 100\,MHz \rightarrow BW = 10\,MHz$

$f(t) = \cos 2\pi f_c t \rightarrow f_2 = 2\,GHz$

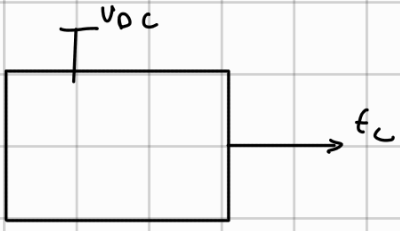
oluşacak frekanslar 2.1 GHz ve 1,9 GHz

Osilatörler & VCO (Voltage controlled oscilator)



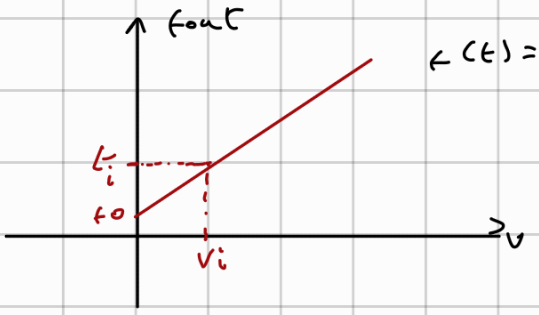
* frequency stability, phase Noise

Osilatör



kristal osilatörlerin beslemeye ihtiyacı yoktur

VCO : gerilim kontrollü osilatör

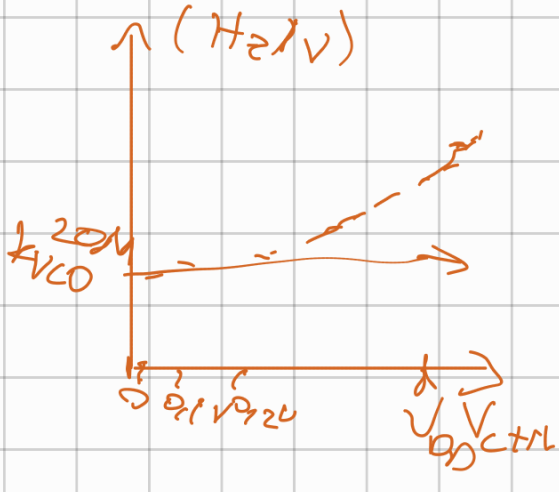


$f(t) = f_o + k_o v_{in}(t)$

$= Hz/V$

$\downarrow$

K_{VCO}

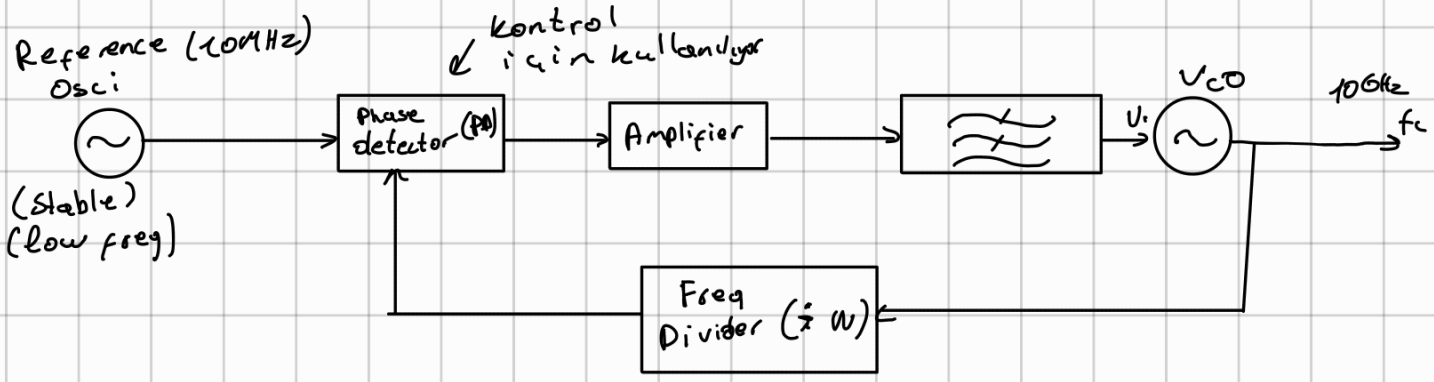


Osilatör çeşitleri → VCO

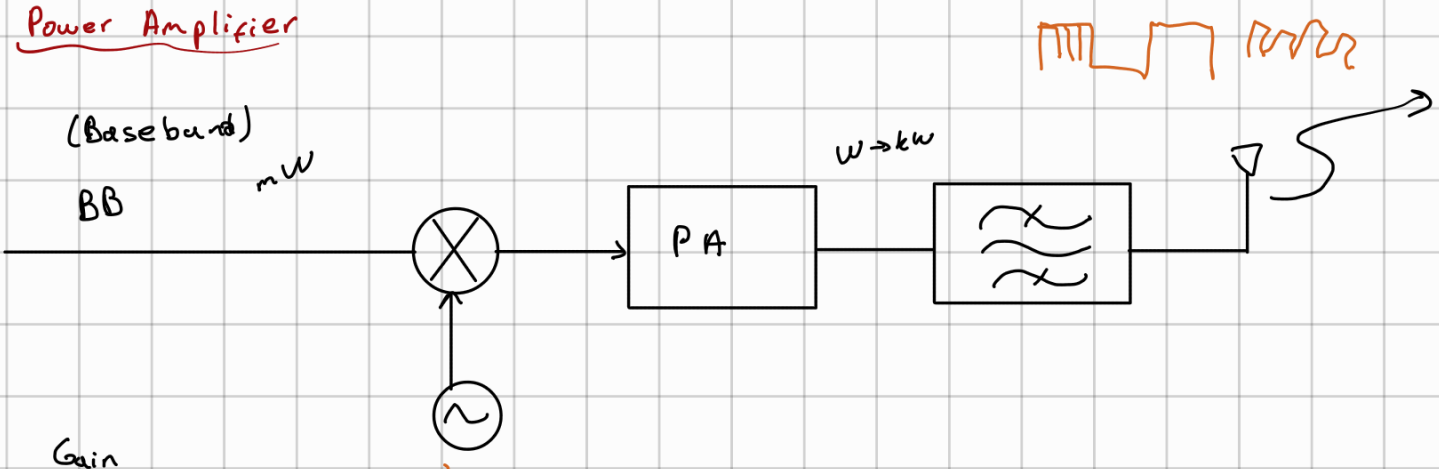
→ TCXO (Temperature controlled Osc)
↳ sıcak kontrollü, geri besleme ile sıcaklık etkisini minimize eder

↳ OCXO (Oven Controlled Osc)
fırın kontrollü

PLL (Phase Locked loop)



Power Amplifier



Gain

(Max) Output power

Bandwidth

Power efficiency (PAE)
Power added efficiency

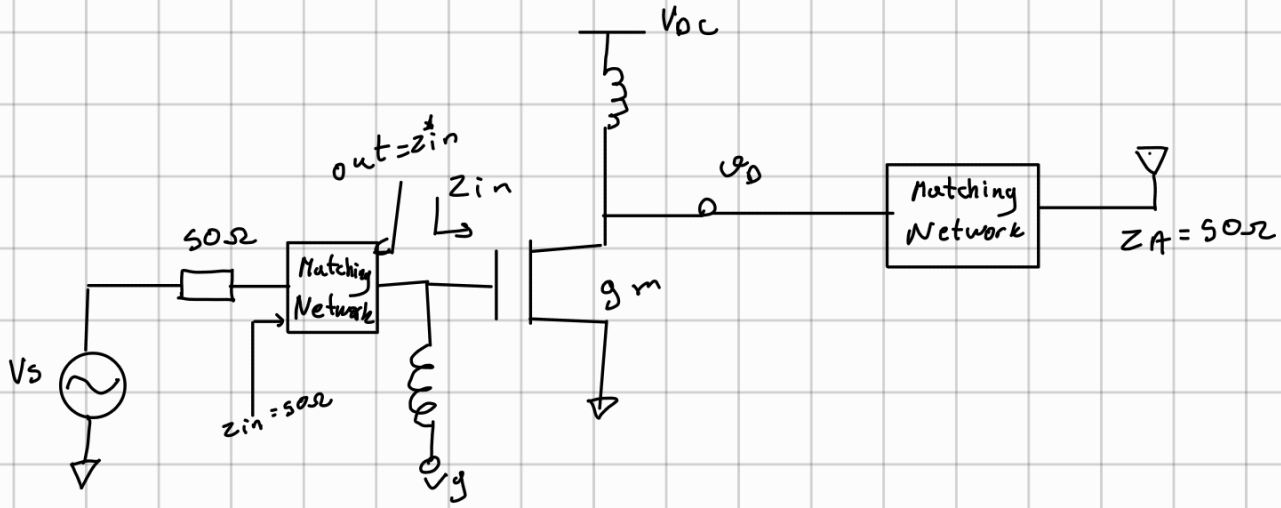
Linearity

I/O Matching

A, AB, B, C → Linear

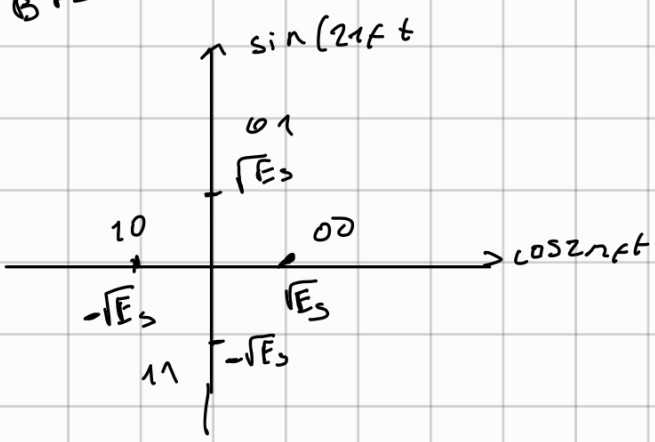
F, E → High power efficiency

D → Low freq

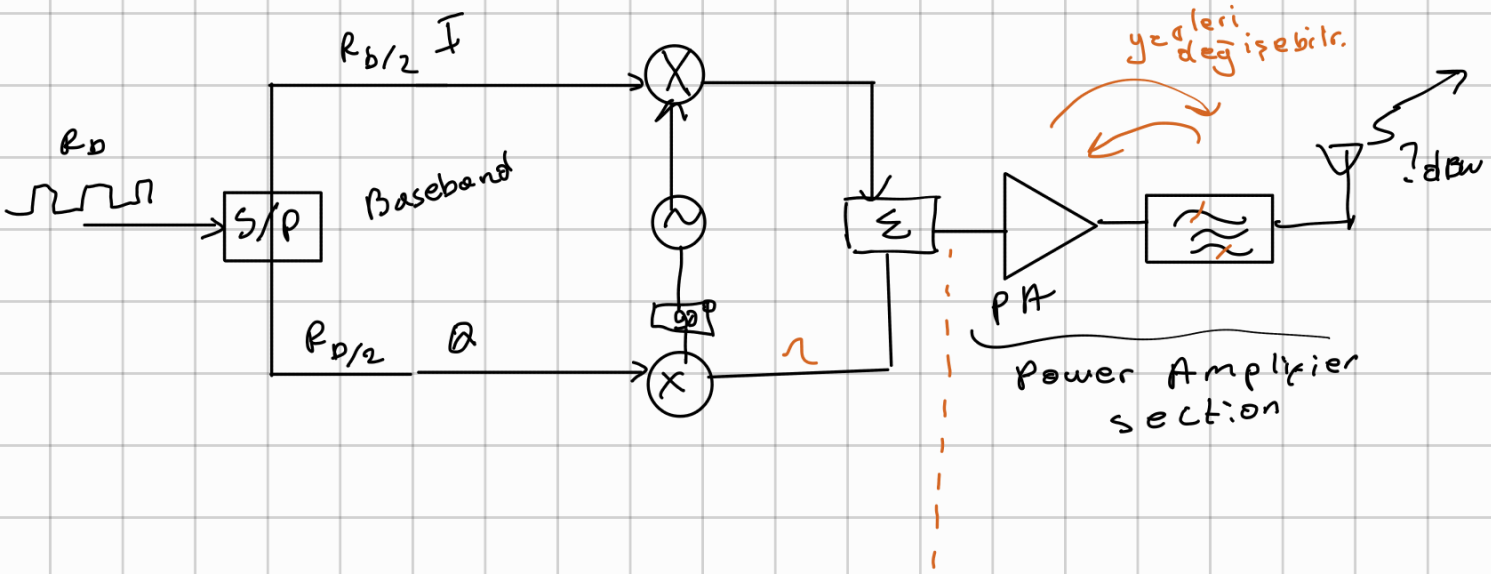


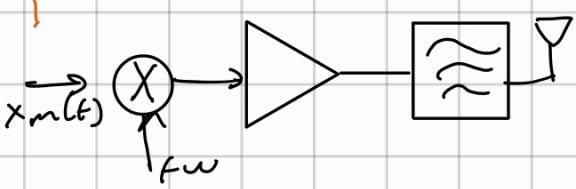
Modulation

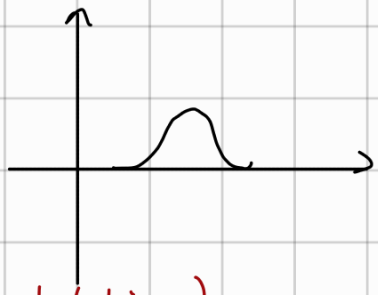
BPSK:
'0' → $\sqrt{E_s}\cos 2\pi f t$
'1' → $-\sqrt{E_s}\cos 2\pi f t$



QPSK: '0 0' → $\sqrt{E_s}\cos 2\pi f t + \sqrt{E_s}\sin 2\pi f t$
'0 1'
'1 0' →
'1 1'

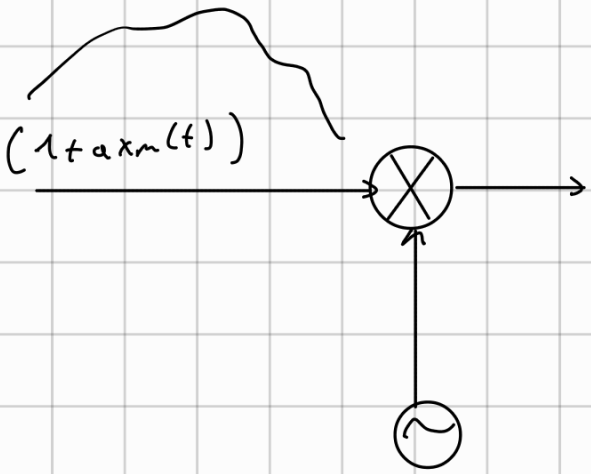


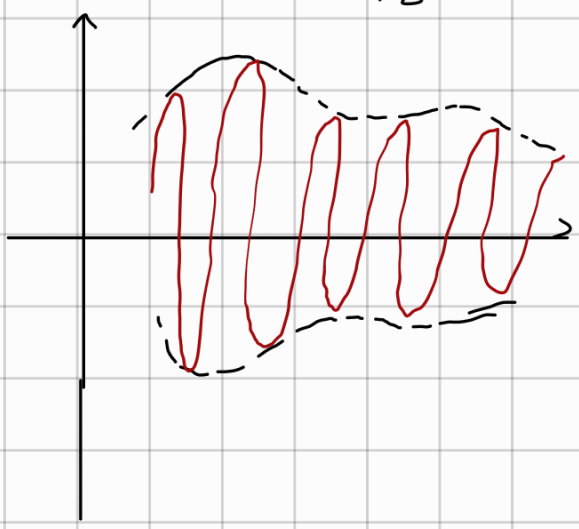




AMC Amplitude Modulation)

high Freq RF signal



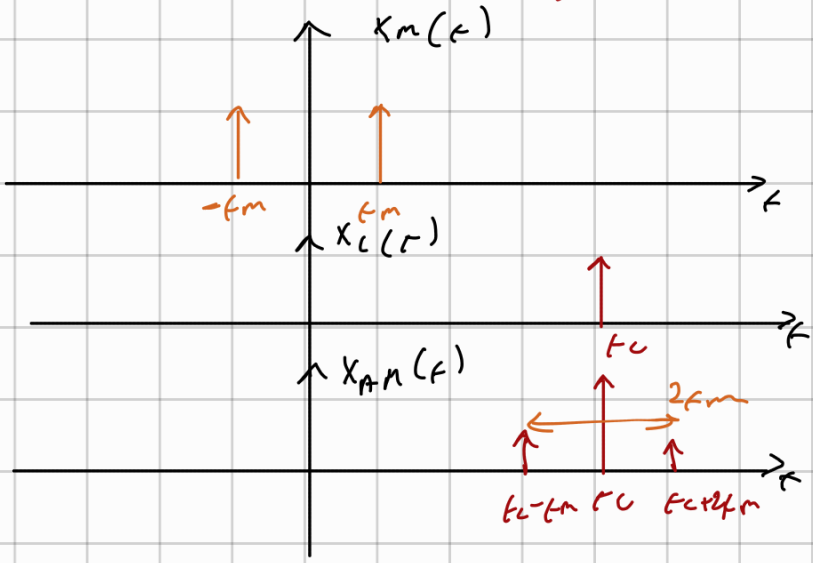




$$x_{am}(t) = [1 + a\,x_m(t)]\,A_c \cos(2\pi f_c t + \phi_c)$$

normalizasyonsuz

$= A_m \cos(2\pi f_m t + \phi_m)$ olsun



Frequency Modulation (FM)

$$x_{FM}(t) = A_c \cos\left[w_c(t) \cdot t + \Phi_c \right]$$

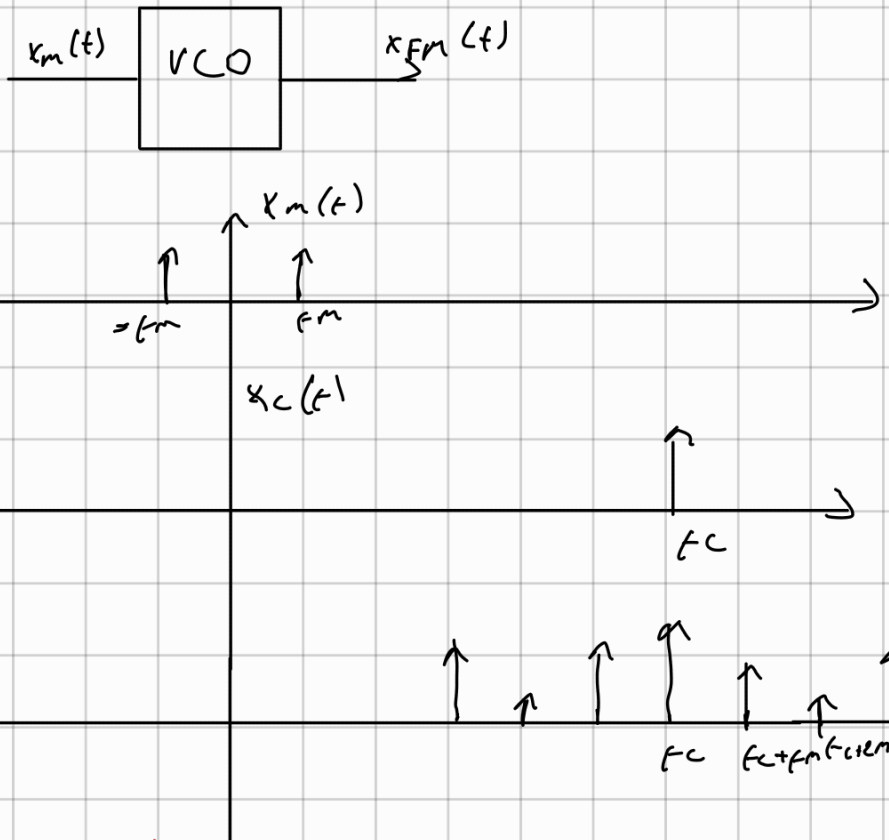
$$w_i(t) = w_c + k_f \cdot A_m \sin(2\pi f_m t)$$

$$BW_{FM} = 2(1 + k_f) f_m$$

$k_f \ll 1 \quad BW_{FM} = 2 f_m$ → NBFM

$$k_f = \frac{f_m}{f_c} \qquad \frac{3\,kHz}{1\,MHz}$$

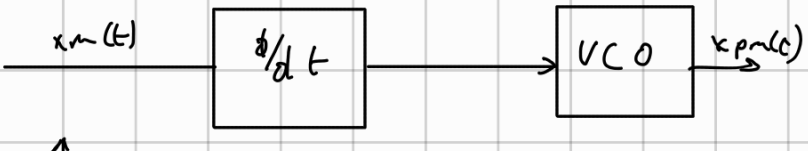
88 - 108 MHz radyo

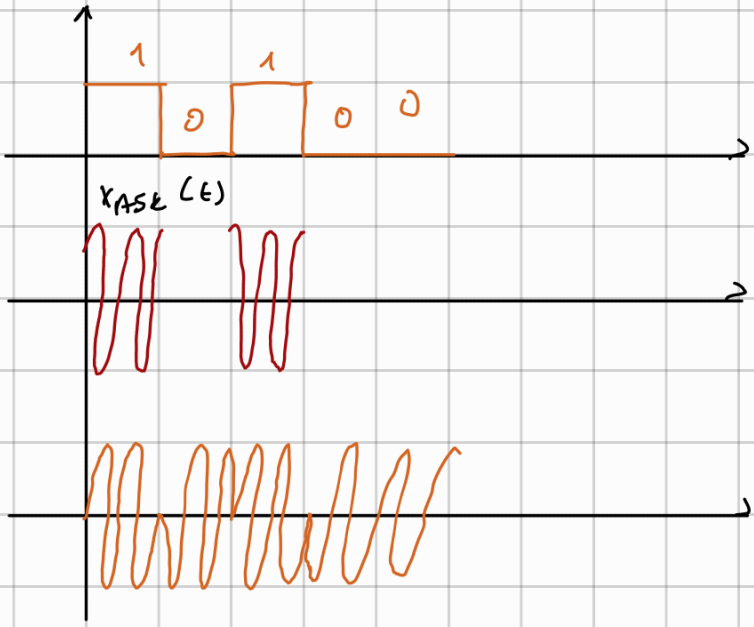


Phase Modulation

$$x_{PM}(t) = A_c \cos\left(w_c t + x_m(t)\right)$$

$$x_m(t) = A_m \cos 2\pi f_m t$$



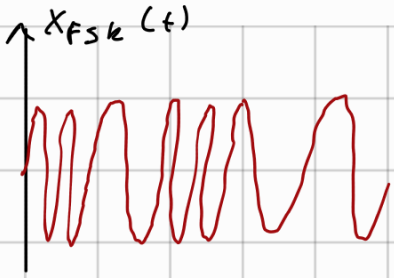


'0' → 0

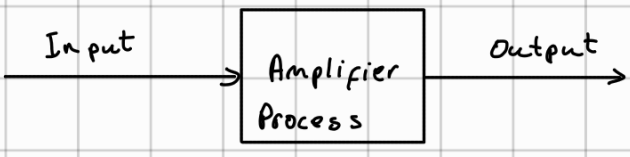
'1' → cos 2πfct

'0' → cos 2πfct

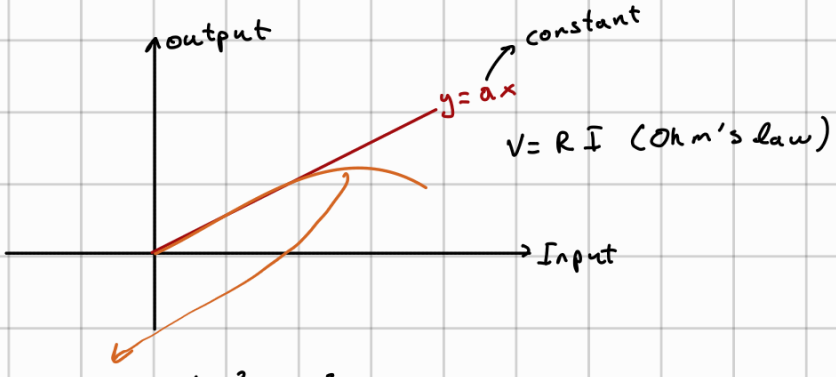
'1' → cos(2πfct + 180°)



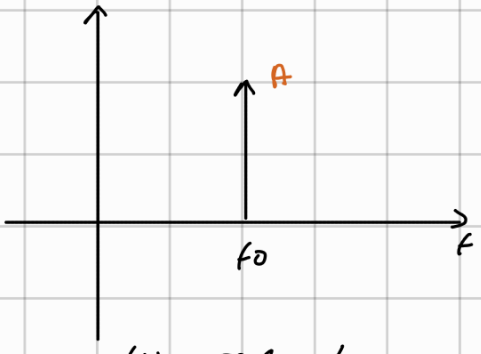
Linearity



$y = a\ x$

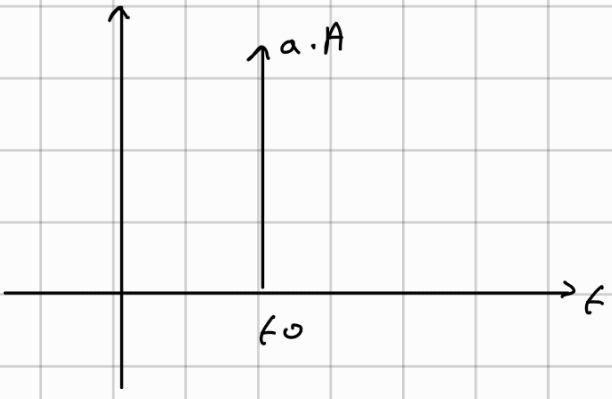


$y = ax + bx^2 + cx^3 + \cdots$

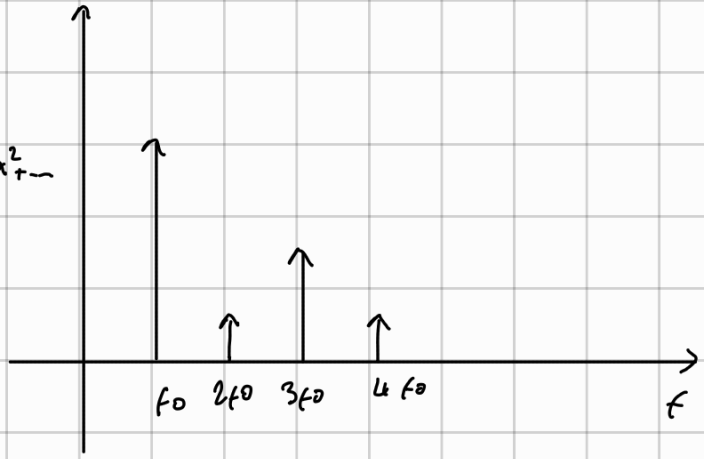


$x(t) = \cos 2\pi f_0 t$

$y = ax$



$y = ax + bx^2 + \cdots$



Nonlinearity Problems

- Gain Compression
- Intermodulation
- Desensitization

Phasor ★5

$$x(t) = A\cos(\omega t+\phi) = Re\{A\cos(\omega t+\phi) + j\sin(\omega t+\phi)\}$$

$$X = Ae^{j\phi} \qquad = Re\{Ae^{j(\omega t+\phi)}\}$$

$$= Re\{A\cdot e^{j\omega t}e^{j\phi}\} = Re\{e^{j\omega t}\cdot X\}$$

$$v(t) = 2\cos(\omega t+30^\circ) \rightarrow V = 2e^{j30}$$

Impedance

Admitance : $Y = \frac{1}{Z}$

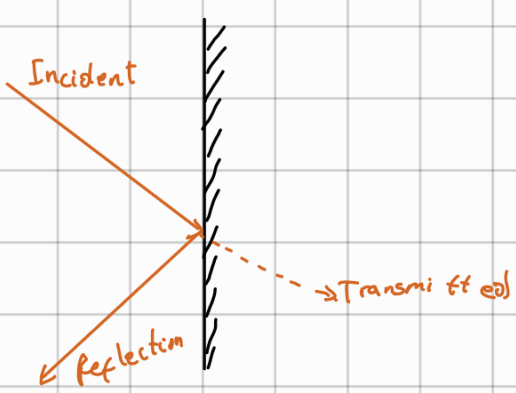
$$R = \frac{V}{I} \Rightarrow Z(f) = \frac{V(f)}{I(f)} = R + jX$$

$R \rightarrow \quad Z = R$

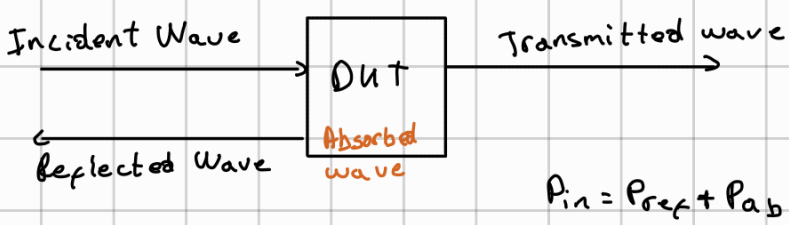
$L \rightarrow \quad Z_L = j\omega L$

$C \rightarrow \quad Z_L = \frac{1}{j\omega C}$

Reflection, Transmission & matching

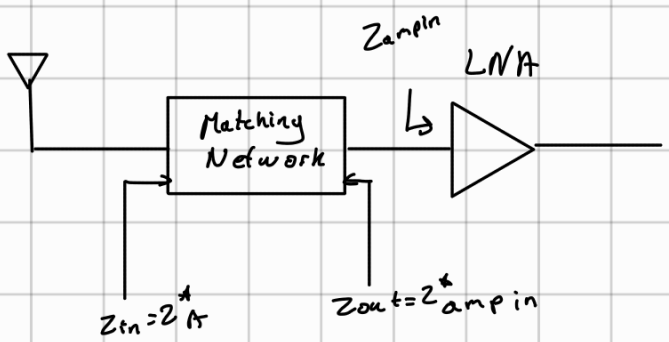


Reflection = γ_{in}



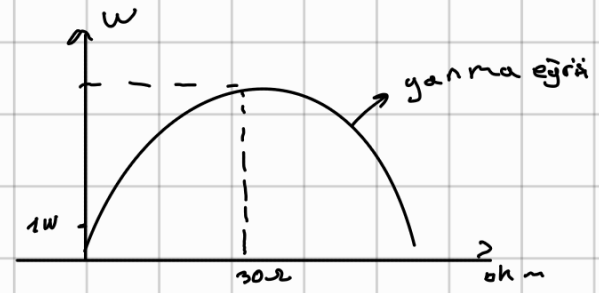
$P_{in} = P_{ref} + P_{ab}$ ← Transmitted wave

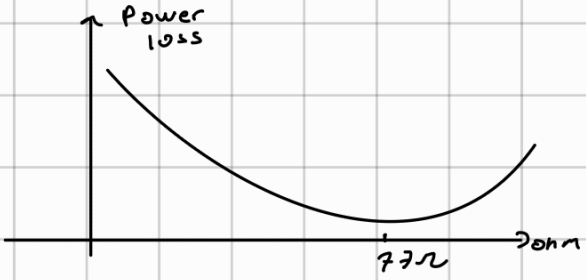
Matching



50 Ω

Powe handling 30Ω

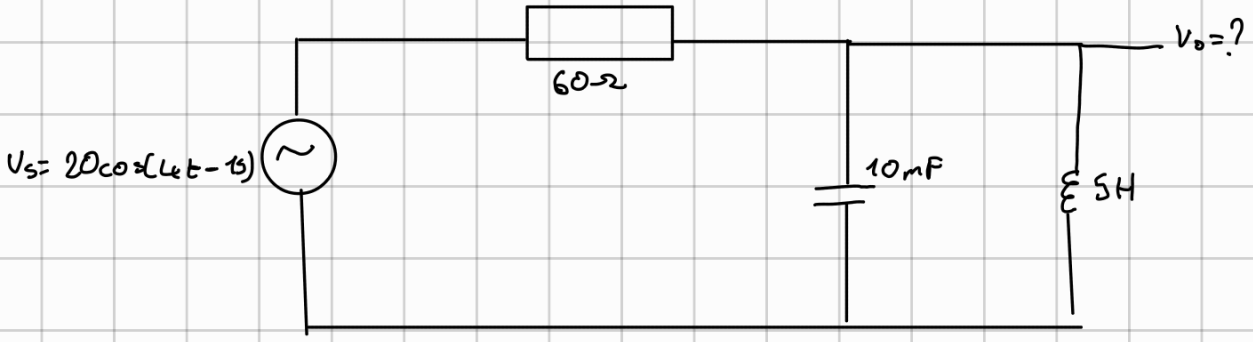




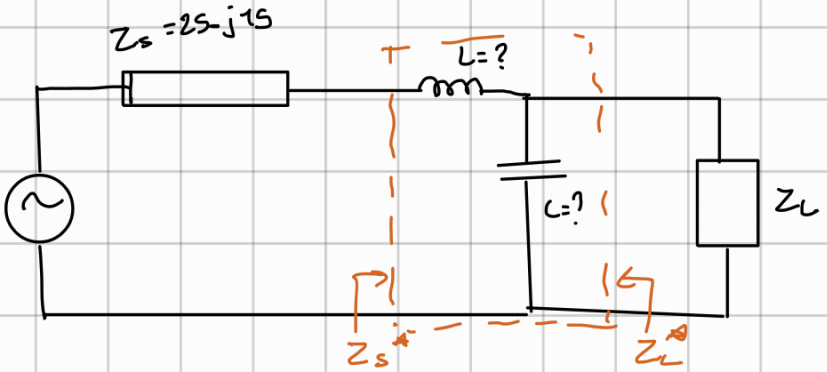
53,5 Ω → 50Ω

5.dev

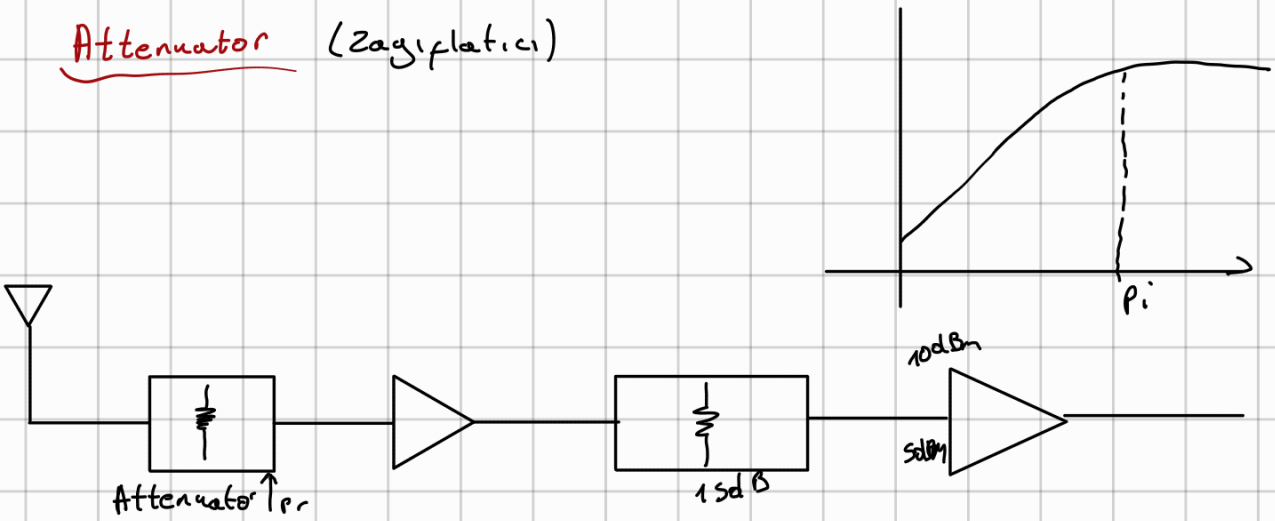
①



②



Attenuator (Zayıflatıcı)



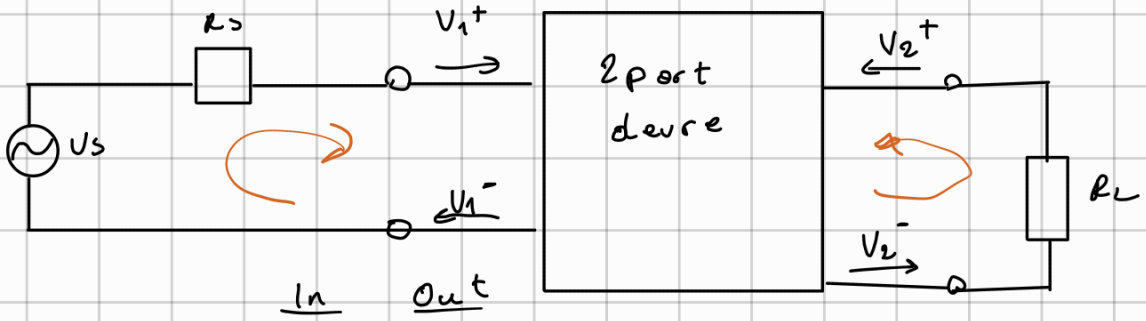
Step Attenuator 1dB → 0 ~ 30dB

2dB → 0 ~ 40dB

Amfiyi doyurmadan linear bölgede çalışması için

Fixed Attention

S-parameters



	In	Out
Incident Wave	V_1^+	V_2^+
Reflected Wave	V_1^-	V_2^-

$$\begin{bmatrix} V_1^- \\ V_2^- \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} V_1^+ \\ V_2^+ \end{bmatrix}$$

$$V_1^- = S_{11} V_1^+ + S_{12} V_2^+$$

$$V_2^- = S_{21} V_1^+ + S_{22} V_2^+$$

$$S_{11} = \left. \frac{V_1^-}{V_1^+} \right|_{V_2^+ = 0}$$

shows the accuracy of input matching

V_1^- nekadar küçükse yansıma o kadar azdır

$S_{11} \leq -15$ dB den küçükse

Shows the accuracy of output matching

$$S_{22} = \frac{V_2^-}{V_2^+}\Big|_{V_1^+=0}$$

$S_{22} \leq -15dB$

$$S_{12} = \frac{V_1^-}{V_2^+}\Big|_{V_1^+=0}$$

Reverse Isolation

çıkıştan girişe gelen sinyal

$$S_{21} = \frac{V_2^-}{V_1^+}\Big|_{V_2^+=0}$$

Gain